



CAMBRIDGE INTERNATIONAL AS & A LEVEL 9702

# AS & A2 Physics — Topic Outline

Topics 1–11 form the AS year; topics 12–25 form A2. Both years are taught with live core sessions, recorded modules, topic quizzes and timed past papers.

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## AS LEVEL — TOPICS 1–11

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| <b>1</b>  | <b>Physical quantities and units</b><br>SI units · errors and uncertainties · scalars and vectors |
| <b>2</b>  | <b>Kinematics</b><br>Equations of motion · motion graphs · projectiles                            |
| <b>3</b>  | <b>Dynamics</b><br>Newton's laws · momentum and impulse · collisions                              |
| <b>4</b>  | <b>Forces, density and pressure</b><br>Turning effects · equilibrium · density and pressure       |
| <b>5</b>  | <b>Work, energy and power</b><br>Energy conservation · gravitational and elastic PE · efficiency  |
| <b>6</b>  | <b>Deformation of solids</b><br>Stress, strain and the Young modulus                              |
| <b>7</b>  | <b>Waves</b><br>Progressive waves · transverse and longitudinal · Doppler · EM spectrum           |
| <b>8</b>  | <b>Superposition</b><br>Stationary waves · diffraction · two-source interference · the grating    |
| <b>9</b>  | <b>Electricity</b><br>Current, potential difference, resistance and resistivity                   |
| <b>10</b> | <b>D.C. circuits</b><br>Series and parallel · potential dividers · e.m.f. and internal resistance |
| <b>11</b> | <b>Particle physics</b><br>Atoms, nuclei and radiation · fundamental particles                    |

## A2 LEVEL — TOPICS 12–25

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| <b>12</b> | <b>Motion in a circle</b><br>Radians · angular speed · centripetal acceleration and force |
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**13      Gravitational fields**

Field lines · Newton's law of gravitation · orbits · potential

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**14      Temperature**

Thermal equilibrium · scales · specific heat and latent heat

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**15      Ideal gases**

The mole · equation of state · kinetic theory

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**16      Thermodynamics**

Internal energy · the first law of thermodynamics

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**17      Oscillations**

SHM · energy in SHM · damping and resonance

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**18      Electric fields**

Field strength · Coulomb's law · potential

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**19      Capacitance**

Capacitors in circuits · energy stored · discharge through a resistor

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**20      Magnetic fields**

Force on a conductor and on a charge · flux · electromagnetic induction

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**21      Alternating currents**

Characteristics · rectification · transformers

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**22      Quantum physics**

Photons · photoelectric effect · wave-particle duality · energy levels

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**23      Nuclear physics**

Mass defect and binding energy · radioactive decay

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**24      Medical physics**

Ultrasound · X-rays · PET scanning

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**25      Astronomy and cosmology**

Standard candles · stellar radii · Hubble's law

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**ASSESSMENT**

**Paper 1** — Multiple choice (AS).

**Paper 2** — AS structured questions.

**Paper 3** — Advanced practical skills.

**Paper 4** — A2 structured questions.

**Paper 5** — Planning, analysis and evaluation.

**Paper 5 is a technique, not a knowledge paper.** It rewards a repeatable method, which is why it gets its own weekly slot rather than a few sessions before the exam.